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Ambient Humidity Influence on CO Detection with SnO_2 Gas Sensing Materials. A Combined DRIFTS/DFT Investigation

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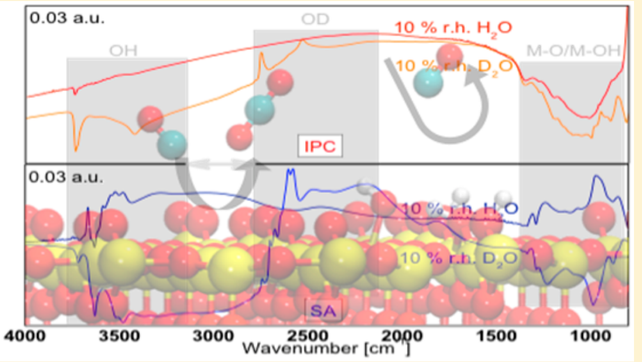
ABSTRACT: Diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) and first-principles calculations are performed to investigate the different ways in which water reacts with a SnO_2 surface and to evaluate the cross-interference of humidity on the detection of CO. Two different materials, chosen because of their very different properties, are investigated. The experimental results are interpreted with the help of theoretical modeling of two clean and defective surfaces, namely, (110) and (101). The experimental results show, and the theoretical calculations confirm, that water vapor can interfere with the CO detection in different ways depending on the active surface and the concentration of oxygen vacancies. This is related to the different ways in which the water vapor reacts with tin oxide: on the one hand, it can reduce the (101) surface; on the other hand, it can heal the oxygen vacancies of the defective (110) surface.



INTRODUCTION

More than 60 years ago Heiland et al.[1,2] showed that the conductivity of semiconducting metal oxides (SMOX) (in that case, single crystalline ZnO) depends on the composition of the surrounding atmosphere. More than 50 years ago Seiyama et al.[3] proposed the coupling of SMOX with gas chromatographs (ZnO thin film). The first gas sensors based on SnO_2 were realized by Taguchi,[4] who later founded Figaro Engineering, the first company to commercialize SMOX based gas sensors worldwide. Due to their inherent advantages (high sensitivity, low cost, potential for miniaturization, good stability), SMOX based sensors have been used in various applications, e.g., explosive and toxic gas alarms, controls for air intake into car cabin, monitors for industrial processes, etc.[5] Most of the SMOX gas sensors are based on porous layers of SnO_2 and WO_3, because these oxides meet the application specific requirements of sensitivity and stability[6−8] and, currently, the drive is toward further miniaturization and increased selectivity and sensitivity to make possible their integration in a host of mobile and household devices and help make possible the Internet of Things (IoT). Gas sensing with semiconducting metal oxides (SMOX) based sensors in ambient conditions means that a changing humidity background needs to be taken into account. Considering how important this issue can be for practical applications, it is surprising that not too many publications focus on that. Generally, the publications dealing with humidity effect and SnO_2, which is considered to be the prototype SMOX for gas sensing, report a resistance decrease under humidity exposure at an operation temperature of 300 °C.[9−11] This implies that H_2O acts as a reducing gas, when reacting with SnO_2 surface.

The reducing effect is still best described with the reaction mechanisms proposed by Heiland and Kohl[12] in 1988:

- The first mechanism includes the homolytic dissociation of H_2O and its reaction with one lattice oxygen, O_O, to form one terminal hydroxyl group, $(Sn_{Sn}{}^{+}{-}OH^{-})$, at a tin place, Sn_{Sn}, and one rooted hydroxyl group, $(OH)_O{}^{+}$, which reacts as a surface donor freeing one electron, e^{-}, to the conduction band. The reaction can be described by the following eq 1:

$$H_2O + Sn_{Sn} + O_O \rightleftharpoons (Sn_{Sn}{}^{+}{-}OH^{-}) + (OH)_O{}^{+} + e^{-} \quad (1)$$

- The second proposed mechanism includes also a reaction of water with lattice oxygen forming two terminal hydroxyl groups, $(Sn_{Sn}{}^{+}{-}OH^{-})$, and one oxygen vacancy, $V_O{}^{++}$. The formation of the latter is providing two electrons to the conduction band. The reaction is best described by the following eq 2:

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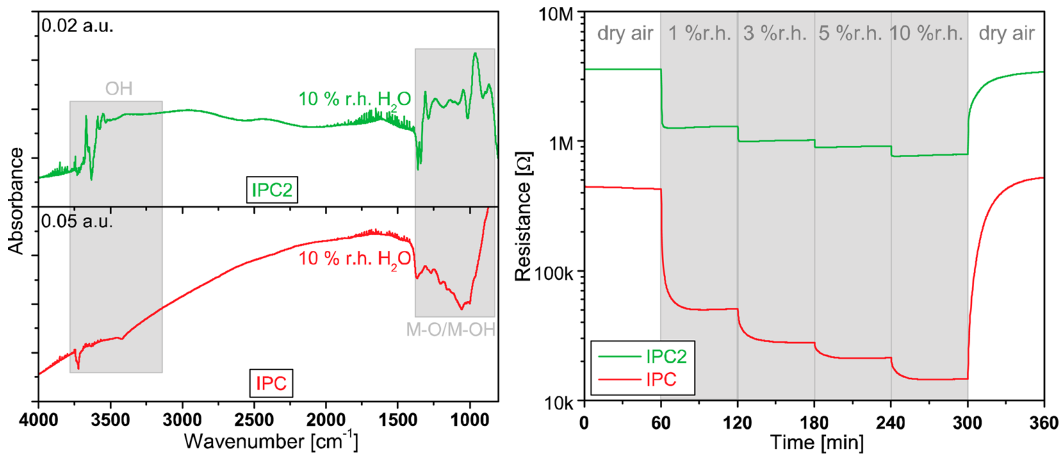

Figure 1. Left: DRIFTS absorbance spectra of IPC and IPC2 operated at 300 °C under exposure of 10% relative humidity (r.h.) H_2O @ 25 °C. Reference spectra: dry syn. air. Right: corresponding resistance measurements with 1, 3, 5, and 10% r.h. H_2O @ 25 °C in dry syn. air background. Adapted from ref 13.

$$H_2O + 2Sn_{Sn} + O_O \rightleftharpoons 2(Sn_{Sn}{}^{+}\text{–}OH^{-}) + V_O{}^{++} + 2e^{-} \quad (2)$$

So, the formation of hydroxyl groups is linked to the reducing effect of water on SnO_2, and therefore the formation of hydroxyl groups under water vapor exposure should be directly correlated to a decrease in resistance of SnO_2. However, as already published DRIFTS spectra and resistance data show,[13] the reality is more complicated: the effect of humidity on two in-house prepared materials show that although significantly more hydroxyl groups are formed on one material, IPC2, than the other, IPC, an opposite electrical effect is observable (Figure 1).

The preparation of the two materials presented in Figure 1 is described in detail elsewhere.[13] It is important to note that they were both synthesized by an aqueous sol–gel route starting from tin chloride; the only difference is the calcination temperature, which was 1000 °C for IPC and 450 °C for IPC2.

Here, with the aim to clarify this apparent contradiction, further investigations were performed on two selected SnO_2 materials by assessing the sensor performance and using in operando DRIFT spectroscopy; the selection criterion was the very different humidity related gas sensing behavior. One of the materials is prepared in house (IPC); the other is based on a commercially available powder (SA). A theoretical study based on first-principles calculations is associated with the experimental investigation to detail the adsorption of isolated H_2O and CO molecules on SnO_2 surfaces and the effect of the humidity on CO detection. Carbon monoxide was chosen as a model gas for the understanding of the well documented interfering effect of humidity on the detection of target gases.[13,14–17]

EXPERIMENTAL METHODS

Material Synthesis. *IPC.* The in-house prepared SnO_2 powder is gained through an aqueous sol–gel process starting with a $SnCl_4$ solution. An ammonia solution is added dropwise. During this process both solutions are kept at 0 °C and stirred thoroughly to ensure homogeneous and small precipitates of $Sn(OH)_4$. The gained gel is washed several times with distilled water until a pH value of 7 is achieved. It is then dried at 120 °C and finally calcined in a tube furnace (Heraeus ROK 6/30) for 8 h at 1000 °C (IPC). A more detailed description of the synthesis and characterization of the material can be found in refs 18 and 19.

SA. A SnO_2 powder (Sigma-Aldrich, <100 nm particle size (BET)) is suspended in deionized water and a stock solution of deionized water and HCl is added. The whole suspension is stirred for 48 h, then dried in an oven at 80 °C (Heraeus UT12), and afterward calcined at 450 °C for 1 h in a tube furnace (Heraeus ROK 6/30).

Sensor Fabrication. The sensitive layer of the sensor is applied onto the substrate by screen printing a paste of 1,2-propanediol (Sigma-Aldrich, 99.5+% ACS reagent) and the corresponding SnO_2 powder. The mixture is ground with mortar and pestle until a homogeneous viscous paste is achieved. During the screen printing process the as-prepared paste is pressed through the screen printing mask with a rubber squeegee onto Al_2O_3 substrates. The sensors are left to settle at room temperature for 1 h and dried in an oven (Heraeus UT12) at 80 °C overnight. In the last step the sensors are annealed by heating them for 10 min each at 400–500–400 °C with two steps to cool down by moving it through a tube furnace (Heraeus ROK 6/30).

dc Resistance Measurements. The changes of the sensing layer resistance are measured with a Keithley 199 electrometer. A computer controlled gas mixing system equipped with data acquisition cards and mass flow controllers provides various concentrations of the test gas CO as well as different humidity backgrounds (0% relative humidity (r.h.) to 50% r.h. H_2O @ 25 °C). The sensors are measured at an operation temperature of 300 °C. The sensor signal is defined as the ratio of two resistances:

$$S = R_{\text{reference gas}}/R_{\text{test gas}}$$

DRIFT Spectroscopy. The DRIFTS measurements are performed with an evacuated FT-IR spectrometer (Bruker V80v), which is equipped with a Harrick cell ("Praying Mantis"). In this cell a special sensor DRIFTS chamber is mounted. The spectrometer consists of an interferometer and a KBr window (beam splitter), and an infrared light is recorded with a LN-MCT-Narrow 24 h photodetector (Liquid nitrogen cooled-mercury cadmium telluride). The spectra are recorded every 15 min with a resolution of 4 cm^{-1} and 1024 sample scans. The apodization function Blackman-Harris-3-Term is

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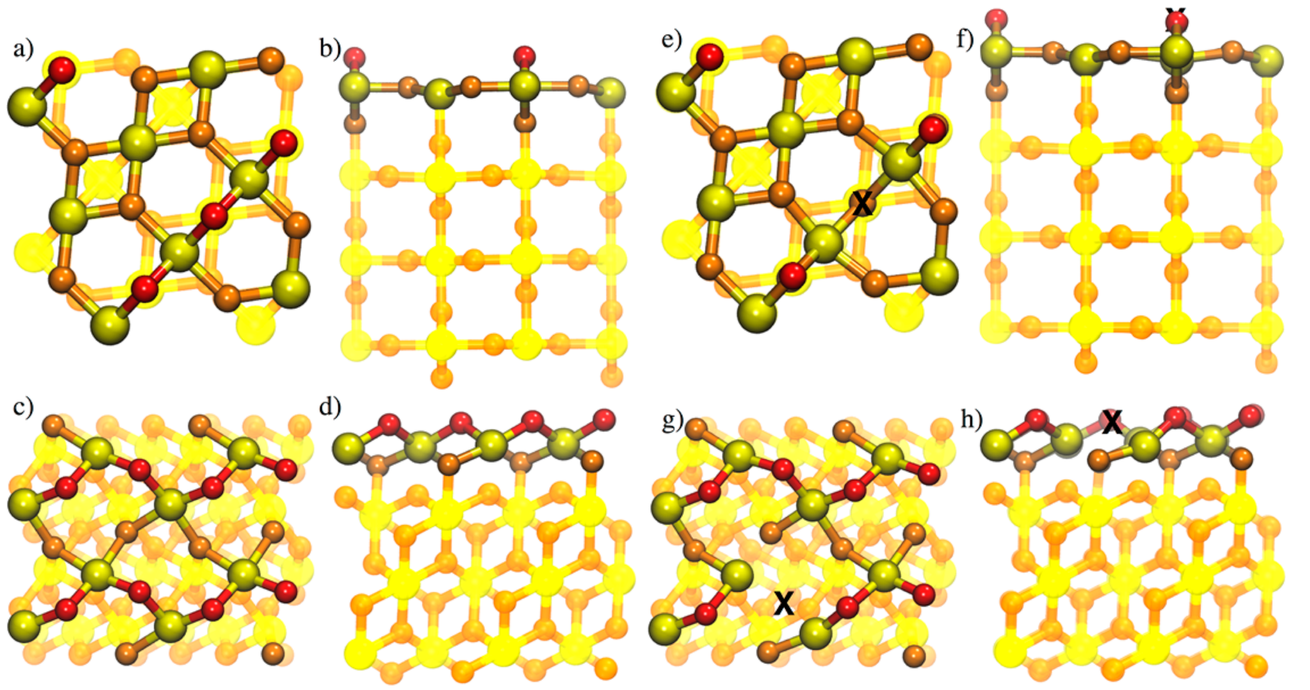

Figure 2. (a) Top and (b) side views of the (110) surface. (c) Top and (d) side views of the (101) surface. (e) Top and (f) side views of the (110) surface containing an O_O vacancy. (g) Top and (h) side views of the (101) surface containing an O_O vacancy. The black crosses on (e) to (h) indicate vacancy, where the O_O atom has been removed from the topmost layer of the SnO_2 substrates. Topmost layers of atoms are highlighted, whereas the layers below the surface are shadowed. Tin atoms are in yellow. Surface out-of-plane oxygen atoms are highlighted in red; other oxygen atoms are in orange. The color scheme and surface representation are used throughout the paper.

applied and the aperture was set to 3.5 mm to ensure that the DRIFTS measurement is performed only on the sensitive layer. The evaluated absorbance spectra are calculated by dividing two single channels through one another:

$$\text{absorbance} = -\log(\text{sample spectrum}/\text{reference spectrum})$$

For a better understanding of the procedure, see Figure S1.

In parallel to the DRIFTS measurements, the resistance changes of the mounted sensor are recorded with a Keithley 617 electrometer. The sensor is heated to 300 °C, and a computer controlled gas mixing system, which is equipped with mass flow controllers and a data acquisition card, provides the DRIFTS setup with the relevant test gases (^{12}CO, ^{13}CO, H_2O, D_2O) and gas concentrations.

Raman Spectroscopy. The Raman spectra of the two sensors were recorded using a Renishaw inVia Reflex Raman spectrometer with an excitation wavelength of 532 nm.

THEORETICAL METHODS

Details of Calculations. The periodic calculations were performed using the DFT method based on the GGA approximation employing the PBE exchange−correlation functional as implemented in the plane-waves program VASP.[20−22] Single-point computations, from relaxed configurations, were performed following the scheme proposed by Heyd, Scuseria, and Ernzerhof (HSE), which separates the exchange energy into short-range and long-range components.[23−27] The projector-augmented wave (PAW) potentials were applied for the core electron representation.[28,29] A converged value of E_{cut} = 400 eV was used for the cutoff energy of the plane wave. The integration in reciprocal space was performed with a Monkhorst−Pack grid.[30]

Two simulations cells are used. The stoichiometric SnO_2 (101) and (110) surfaces are modeled by periodic slabs depicted in Figure 2.[31−34] For the (101) surface, the cell is composed of a total number of 96 atoms with 32 tin atoms and 64 oxygen atoms corresponding to a cell dimension of 11.42 × 9.47 × 25.00 Å^3 with a surface area of 108.19 Å^2. For the (110) surface, the cell was composed of a total number of 96 atoms with 32 tin atoms and 64 oxygen atoms corresponding to a cell dimension of 9.48 × 9.48 × 25 Å^3 with a surface area of 89.87 Å^2. For both slabs, a vacuum zone of 15 Å in the z direction was used to create a surface effect. Twenty-four Sn and O atoms were frozen in their bulk positions at the bottom of the cell to simulate a bulk effect. All other atoms were free to relax. The periodic slab was repeated in the three directions. A k-point mesh of (2 × 2 x 1) was used. These simulation cell parameters have been checked to avoid any interaction with other periodic cells.

Both (110) and (101) surfaces exhibit two types of oxygen atoms on the surface (Figure 2a−d): there are 2-fold coordinated O atoms called out-of-plane oxygen atoms and denoted as O_O in the following. All other oxygen atoms below the surface layer and deeper into the bulk are 3-fold coordinated.

Defective surfaces, shown as substoichiometric system containing one oxygen vacancy, have been obtained by removing a 2-fold coordinated O_O atom from the surface layer. The corresponding energies of oxygen vacancy formation are 2.30 eV and 2.25 on defective (110) and (101) surfaces, respectively. Other 3-fold coordinated oxygen atoms in the topmost layer exhibit higher energies of oxygen vacancy formation (3.45 and 3.64 eV on defective (110) and (101) surfaces respectively[27,31,32]), so that they were not considered in this work. The energy of oxygen vacancy formation ΔE_0 was

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calculated by using the equation $\Delta E_0 = E_{vac} + {}^1/_2E_{O_2} - E_{surface}$, where E_{O_2} is the energy of the isolated spin polarized O_2 molecule, E_{vac} is the energy of a (110) surface or (101) surface with one oxygen vacancy per supercell, and Esurface is the energy of the stoichiometric surface.

Adsorptions of one water molecule are thus performed considering the topology of such clean and defective (110) and (101) surfaces. The isolated H_2O and CO molecules were relaxed in the same simulation cell as the SnO_2 (101) and (110) slabs. Their adsorption sites were identified by screening the respective topology of the (101) and (110) surfaces. Charges were obtained by calculating the difference of the atomic charges on all the atoms of the oxide before and after adsorption using the Bader charges analysis.[35]

Energetic Considerations. In this paper, the adsorption energies ΔE_{ads} are calculated using the following formula:

$$\Delta E_{ads} = E_{\text{adsorbed molecule}} - E_{\text{surface}} - E_{\text{gaseous molecule}}$$

where $E_{\text{adsorbed molecule}}$ is the total energy of the system with a gaseous molecule adsorbed on the SnO_2 surface, E_{surface} and $E_{\text{gaseous molecule}}$ are the total energies of the relaxed surface and isolated gaseous molecule (either H_2O or CO), respectively. Conventionally, negative values indicate exothermic adsorption.

Activation barriers have been determined along the minimum energy path using a drag method. Here, the system is relaxed; while one coordinate is held fixed along the diffusion pathway, all other coordinates of freedom in the system are relaxed. The maximum energy corresponds to the transition state and determines the activation barrier.

■ EXPERIMENTAL RESULTS

To investigate the impact of changing ambient humidity, two different SnO_2 materials were selected. One was prepared in house, IPC, via a sol−gel route and calcined at 1000 °C, whereas the other one, SA, is based on a purchased SnO_2 material, which underwent wet-chemical pretreatment and a calcination temperature of 450 °C. IPC shows a higher crystallinity, a larger average nanoparticles size (110 nm vs 30 nm) and therefore a lower surface area than SA.[36−38] The measured BET surface areas, by N_2 adsorption by using a Micrometrics ASAP 2020 instrument, are 23 m^2/g for SA and 12.5 m^2/g for IPC. The corresponding REM pictures and XRD spectra are shown in Figure S2.

Humidity Dependence of Baseline Resistance. The humidity influence on the baseline resistance of the two different SnO_2 materials was studied by exposing the sensors to eight humidity levels; ranging from 0.4% to 80% r.h. @ 25 °C (Figure 3).

It is quite obvious that the baseline resistance of IPC is strongly humidity dependent. Already the first increase from 0.1% r.h. (dry syn. air) to 0.4% r.h. @ 25 °C determines a sensor signal of 5. Following increases of humidity, there is a further decrease in the baseline resistance; the corresponding sensor signals span from 5 to 44 (89% r.h. @ 25 °C).

On SA this is not the case; only slight changes with increasing water vapor background are observable. Moreover, whereas low relative humidity levels, up to 1% r.h. @25 °C, lead only to a minimal decrease in resistance, higher humidity levels even cause a slight increase in resistance. All in all the effects of water vapor on the baseline resistance of SA are almost negligible. The sensor signals range from 1.1 to 1.2.

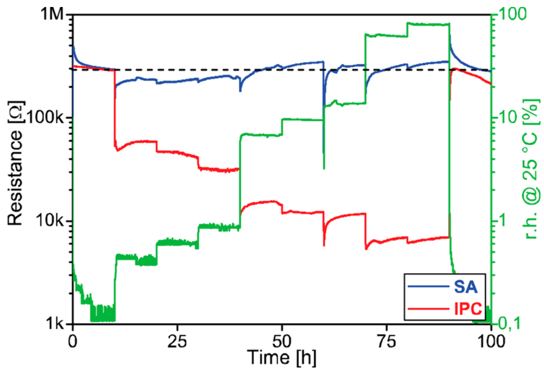

Figure 3. dc resistance measurements of IPC and SA in syn. air background at 300 °C (0.1% r.h. @ 25 °C). The corresponding relative humidity (0.4, 0.6, 0.9, 7, 10, 14, 62, 80% r.h.) @ 25 °C is indicated on the right-hand side *y*-axis.

The observations on IPC show a clear confirmation of the former reports that water vapor reacts as a reducing gas on SnO_2. On the opposite, the findings on the SA do not fit into the current understanding. To get more information about the surface reaction mechanisms, operando DRIFT spectroscopy was used.

In Figure 4, the DRIFT absorbance spectra of IPC and SA are displayed during exposure to 10% r.h. H_2O as well as during exposure to its isotope, heavy water.

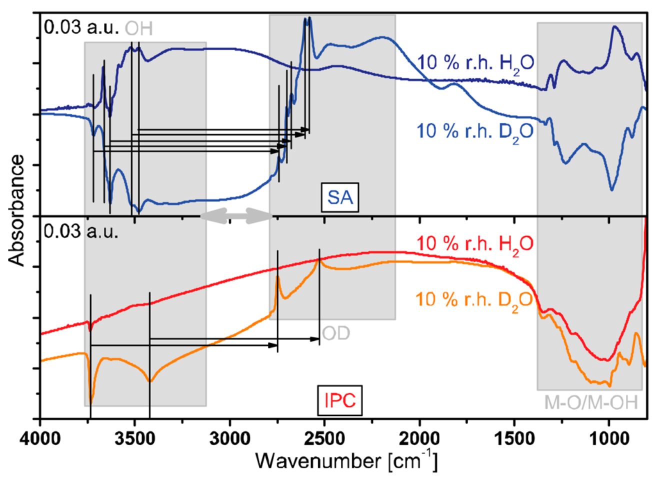

Figure 4. DRIFT absorbance spectra of SA and IPC under 10% r.h. H_2O and D_2O @ 25 °C exposure in syn. air and at an operation temperature of 300 °C. Reference spectra: dry syn. air.

On IPC the formation of OH groups is hardly visible during water vapor exposure. However, a huge decrease in M−O vibration overtones is observable in the fingerprint region 1400−800 cm^{-1}. As the exchange reaction of OH with OD groups during D_2O exposure reveals, also only a few OH groups are preexisting in dry syn. air.

While during H_2O exposure the IPC spectra are mainly defined by the reduction of the material, the spectra of SA are dominated by the formation of OH and M−OH bending vibrations. Also, not only are there a lot of OH groups formed during 10% r.h. H_2O exposure, but also many are already present in dry syn. air, as the exchange with heavy water vapor reveals.

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All in all, the reaction mechanisms of water vapor on SnO_2 proposed by Heiland and Kohl in 1988 are suitable to explain the observations made on IPC with its strong reduction under humidity and the formation of some hydroxyl groups.[12] Additionally, it can be assumed that the surrounding water vapor contributes to the reduction of the material by hindering its reoxidation, described by eq 3.

$$\frac{1}{2}O_2 + V_O^{++} + 2e^- \rightleftharpoons O_O \quad (3)$$

The overall effect of humidity exposure is a decrease of the negative surface charge.

The findings obtained by testing the SA material cannot be similarly explained. The electrical measurements indicate that the surface charge remains unchanged, suggesting that the reaction with water vapor is of an electroneutral nature. This implies that either, if there is a reduction, it is fully compensated by immediate reoxidation or the formation of hydroxyl groups during water vapor exposure has no reducing effect on SA.

Which approach describes the occurring reaction mechanism the best can be verified by removing the possibility of reoxidation by dosing water in the absence of oxygen, i.e., in nitrogen. The corresponding DRIFTS and resistance measurements are shown in Figure S3. It was found that in the absence of oxygen even more hydroxyl groups are formed, while still hardly any electrical effect is observable. These results imply that water does not always react as a reducing gas on SnO_2.

The findings can be best explained by an electrically neutral reaction in which an oxygen vacancy, V_O^{++}, is filled with a rooted hydroxyl group, $(OH)_O^+$, and the residual hydrogen transforms an existing lattice oxygen, O_O, to a second rooted hydroxyl group, $(OH)_O^+$.

$$H_2O + V_O^{++} + O_O \rightleftharpoons 2(OH)_O^+ \quad (4)$$

It means that the reaction with humidity on IPC is mainly dominated by the reduction of the material, whereas an electroneutral reaction mainly dominates on SA. In general, from the experimental results, it can be derived that the ability of a SnO_2 surface to form hydroxyl groups seems to be inversely proportional to the electrical effect humidity has.

Humidity Cross-Influence on CO Sensing. Humidity not only has an effect on the baseline resistance of a SMOX material itself but also contributes to or interferes with the sensor response of other gases, e.g., CO. It is well-known that an increasing water vapor background has a negative impact on the gas sensor signal toward CO on undoped SnO_2.[37,38] This is what is also observable on IPC and SA. In both cases the sensor signal decreases with increasing humidity background from 47/55 to 10/23 (IPC/SA); see Figure S4.

To be able to understand the occurring reactions in more detail, DRIFT spectroscopy was also applied during CO exposure in both dry and humid conditions. The corresponding DRIFT spectra are displayed in Figure 5.

In Figure 5 it can be seen that on IPC both a reduction of the material as well as a decrease in the OH groups is visible during CO exposure in dry syn. air background. Both features are a bit more enhanced in a 3% r.h. H_2O @ 25 °C background. Due to the reducing nature of CO also some CO_2 is formed, in both backgrounds. Its presence in the pores of the material causes the band at 2438 cm^{-1} with its rotational fine structure. The oxidation to CO_2 can be expressed with the following reaction equation:

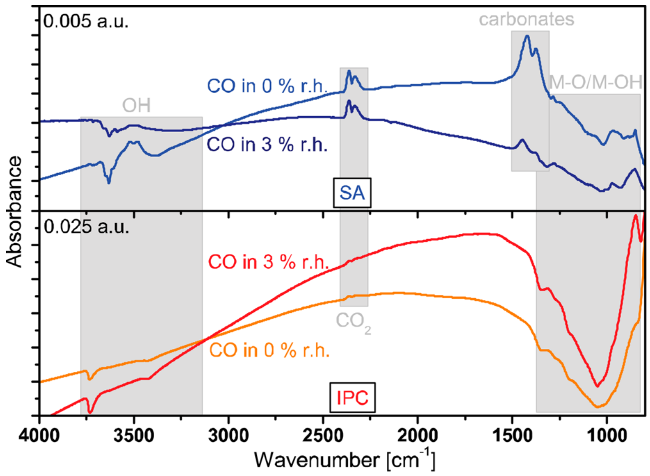

Figure 5. DRIFT absorbance spectra of 500 ppm of CO exposure in dry and in 3% r.h. H_2O @ 25 °C background on IPC and SA at an operation temperature of 300 °C. Reference spectra: dry syn. air and 3% r.h. H_2O @ 25 °C, respectively.

$$CO + O_O \rightleftharpoons CO_{2,gas} + V_O^{++} + 2e^- \quad (5)$$

Although also on SA gaseous CO_2 is detectable, the other features differ. In addition to the CO_2 band, some carbonate bands are visible, which are decreased in the humid background. For the OH bands a shift from terminal to rooted hydroxyl groups occurs in dry conditions, whereas in humid ones several OH groups are removed during CO exposure.

So, on both SnO_2 materials it is obvious that during CO exposure only a weak removal of hydroxyl groups is visible, which implies that CO is hardly able to compete with humidity for reaction partners. This is very much in line with the observation of a decrease in sensor signals to CO in the background of water vapor.

However, if one has a look not to the relative changes (sensor signal), but to the absolute ones, it is obvious from the electrical results that the two SnO_2 materials show opposite trends. For more details see Figure 6.

In the case of the IPC material one observes that the resistance drops not only by increasing the CO concentration but also by increasing the humidity background. On the opposite, for SA the absolute resistance increases under CO exposure when the humidity increases.

Those observations can be explained by taking into consideration the different dominant water vapor reactions for those two materials:

On the one hand, on IPC, both CO and H_2O contribute to the decrease of the resistance and, therefore, their effects are adding up. Both gases react with lattice oxygen, O_O, form oxygen vacancies, V_O^{+2}, and free electrons to the conduction band. The fact that CO and water compete for the same reaction partner, namely, lattice oxygen, is in line with the decrease of the sensor resistance and sensor signal to CO in the presence of humidity.

On the other hand, on SA, the reaction of CO is assumed to follow the same mechanism as on IPC forming CO_2 by reacting with lattice oxygen. However, no direct competition for lattice oxygen takes place between CO and water vapor as the reaction with CO creates oxygen vacancies and water vapor fills them with rooted hydroxyl groups. It means that the presence of humidity seems to hinder the CO sensing in a different way; e.g., some lattice oxygen are replaced by rooted hydroxyl groups

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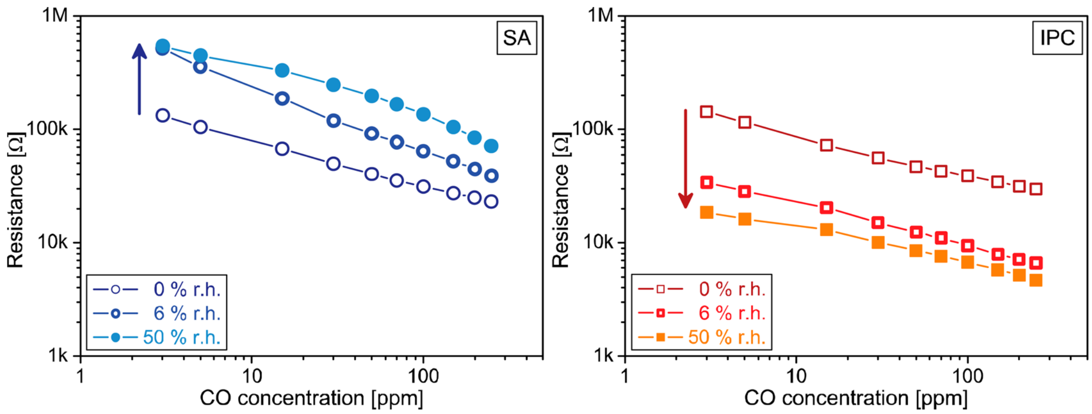

Figure 6. Sensor resistance values for SA (left) and IPC (right) under exposure of 3, 5, 15, 30, 50, 70, 100, 150, 200, and 250 ppm of CO in various humidity backgrounds (0%, 6%, and 50% r.h. @ 25 °C) at an operation temperature of 300 °C. Results of IPC taken from ref 17. Arrows indicate the resistance increase (SA) and decrease (IPC) in increasing humidity backgrounds, respectively.

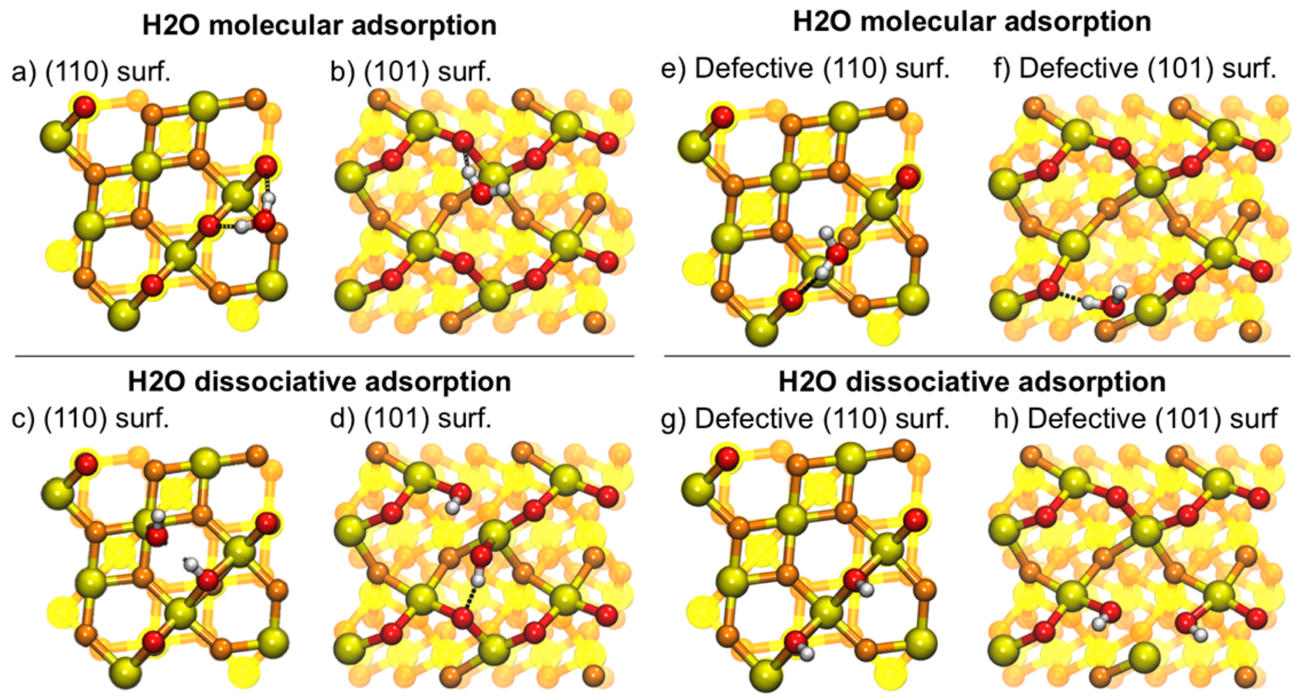

Figure 7. Adsorption modes of H_2O: molecular mode on the clean (a) (110) surface and (b) (101) surface; dissociated mode on the clean (c) (110) surface and (d) (101) surface; molecular mode above a vacancy defect on the (e) (110) surface and (f) (101) surface; dissociated mode above a vacancy defect on the (g) (110) surface and (h) (101) surface. Hydrogen atoms are in white.

and, therefore, not directly available for the reaction with CO. No additive effect on the sensor resistance is observable.

THEORETICAL RESULTS AND DISCUSSION

H_2O Reactions with SnO_2 Surfaces. DFT calculations have been conducted to assess the conclusions of the experiments. Two surface orientations are considered, i.e., (110) and (101), with or without O_O vacancies, on which H_2O and CO are adsorbed.

Both H_2O and CO adsorptions have been widely studied using DFT calculations.[33,39−46] Concerning water adsorption, our findings (Figure 7a) indicate also a physisorbed mode on (110) surface with adsorption energy −0.35 eV, and molecular adsorption modes are reported only on (101) surface (Figure 7b).

Dissociated states are the favored adsorption modes with adsorption energies larger than −1 eV for both surfaces. The dissociation of the H_2O molecule occurs through the adsorption of one OH group on one Sn atom and the adsorption of the dissociated H on one O_O atom[33,40,42−45] (Figure 7c,d).

Clean Surface. On the (110) surface, the spontaneous dissociation of water molecule occurs preferentially on Sn and O atoms, which are first neighbors (Figure 7c) because of the formation of a H-bond between the dissociated H atom coming from the water molecule and the dissociated OH adsorbed on Sn atom (d(H bond) = 1.79 Å).[34,35,43,44] Here the two bonds between Sn−O_O are slightly enlarged, from 2.00 to 2.16 Å but not broken. The corresponding adsorption energy is −1.87 eV and the reaction of water adsorption on (110) can be written as

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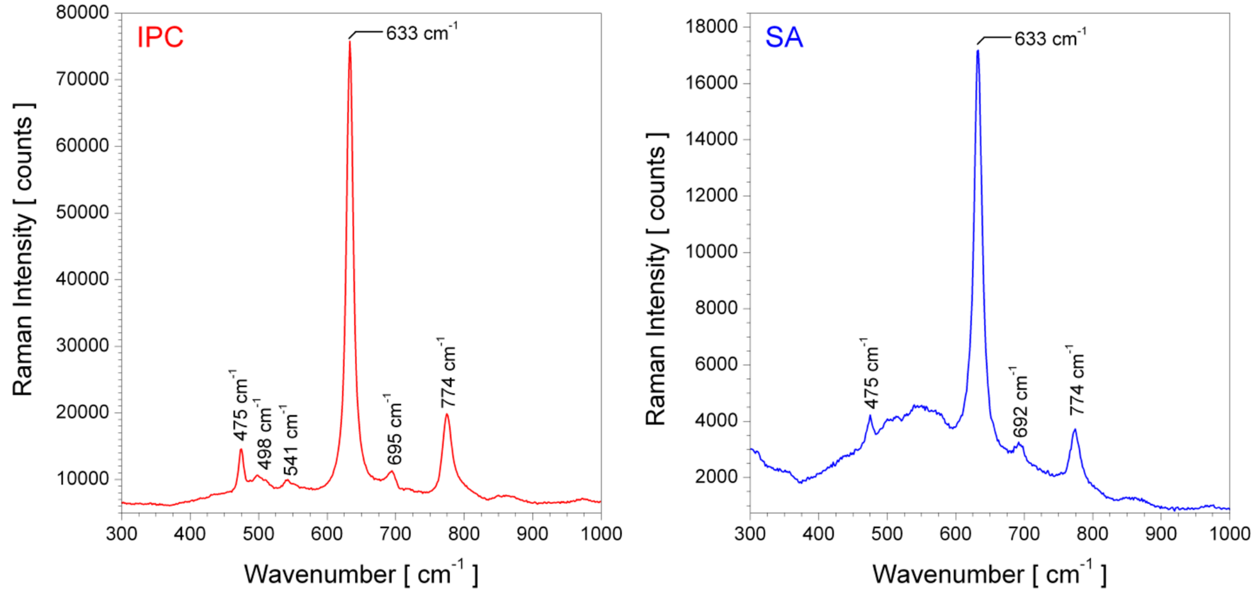

Figure 8. Raman spectra of IPC (left) and SA (right).

$$H_2O_{gas} + Sn_{Sn} + Sn_{Sn}{-}O_O{-}Sn_{Sn} \rightleftharpoons Sn_{Sn}{-}OH + Sn_{Sn}{-}O_OH{-}Sn_{Sn} \quad (6)$$

So, in this case we do not observe the formation of the oxygen vacancies because of the interaction with water vapor. This is supporting the first mechanism proposed by Heiland and Kohl.[12] However, the electrical effect seems to be very limited, −0.12e, when compared to the one correlated with the formation of oxygen vacancies, −1e.

On the (101) surface, the dissociation of water on first neighbors O_O and Sn atom results in two hydroxyl groups (OH). On the opposite to what is observed on the (110) surface, one Sn−O_O bond from the surface breaks (d(Sn−O) = 2.30 Å) when H adsorbs on the O_O atom (Figure 7d indicated by the dashed line in eq 7). The calculated adsorption energy is −1.50 eV and the reaction can be written as

$$H_2O_{gas} + Sn_{Sn} + Sn_{Sn}{-}O_O{-}Sn_{Sn} \rightleftharpoons Sn_{Sn}{-}OH + Sn_{Sn}{-}O_OH\cdots Sn_{Sn} \quad (7)$$

In this case we see the first step in the formation of an oxygen vacancy. We still have the oxygen atom close to the position it would have in the surface lattice and that is the reason we were writing it as $Sn_{Sn}{-}O_OH\cdots Sn_{Sn}$. The electrical effect is still limited, −0.16e, when compared to the formation of an oxygen vacancy but larger than in the case of the real rooted hydroxyl groups formed on the (110) surface.

Defective Surface. In the case of defective (101) and (110) surfaces, the molecular adsorption is enhanced above a vacancy thanks to shorter H-bonds that can be formed between the water molecule and the O_O atoms placed in close vicinity.[34,43] Noteworthy, we observed a molecular adsorption (Figure 7e,f), with calculated adsorption energies of −0.75 and −1.00 eV on (110)-V and (101)-V surfaces, respectively). The dissociative adsorption of the water molecule is still the favored process. It occurs similarly to the atomistic reaction observed on clean surfaces, where OH and H react with Sn and O_O atoms, respectively.

In the case of the (110) surface, the most favorable case is obtained when OH dissociates on the undercoordinated Sn atom, healing the vacancy by re-forming the Sn−O bonds, and the H atom is adsorbed on a distant O_O atom of the surface (Figure 7g). The adsorption energy is larger by 0.39 eV compared to that of the clean surface (−2.26 eV vs −1.87 eV). The healing of the vacancy by the adsorption of OH into V_O is the most favorable scenario for water adsorption on the (110) defective surface and can be summarized as

$$H_2O_{gas} + Sn_{Sn}{-}O_O{-}Sn_{Sn} + Sn_{Sn}\cdots V_O\cdots Sn_{Sn} \rightleftharpoons Sn_{Sn}{-}O_OH{-}Sn_{Sn} + Sn_{Sn}{-}OH{-}Sn_{Sn} \quad (8)$$

The electrical effect of the vacancy healing process on (110) is negligible, −0.07e, which supports the assumption of electroneutrality we made.

In the case of the (101) surface, the most favorable case is obtained when the OH group, resulting from the dissociation of water on the undercoordinated Sn atom, fills the vacancy, and the H atom adsorbs on one of the O_O atom neighboring the vacancy (Figure 7h). For the former process, the adsorption energy is larger by 0.64 eV, when compared to that of the clean surface (−2.14 eV vs −1.50 eV). The Sn−O bond is still broken. For the latter process, when the H atom adsorbs on the O_O atom, a Sn−O_O bond also breaks, similarly to the case of the clean surface. The corresponding reaction can be written as

$$H_2O_{gas} + Sn_{Sn}{-}O_O{-}Sn_{Sn} + Sn_{Sn}\cdots V_O\cdots Sn_{Sn} \rightleftharpoons Sn_{Sn}{-}O_OH\cdots Sn_{Sn} + Sn_{Sn}{-}OH\cdots Sn_{Sn} \quad (9)$$

The electrical effect is a bit larger than in the case of the (110) surface, because of the incipient oxygen vacancy formation, namely, −0.12e, but still very small, when compared to the full formation of the oxygen vacancy.

Our calculations for water vapor adsorption on SnO_2 show that it starts with the dissociation of the water molecule. The resulting hydroxyl group will react differently on clean and defective surfaces. In the first case, it will form a bond to the surface tin atom, i.e., a terminal hydroxyl group. In the second case, it will preferably react with the surface oxygen vacancy

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forming rooted hydroxyl groups. There is a difference between the two surfaces; namely, in the case of (110) the filling of the vacancy is made by forming bonds between the oxygen from the hydroxyl group and the two neighboring tin atoms. In the case of (101), only one bond is formed, between the oxygen and a neighboring tin atom, the second one is broken.

The resulting hydrogen atom always reacts with out-of-plane lattice oxygen atoms for both clean and defective surfaces. There is also a difference between the two surfaces: a bond between a tin atom and the out-of-plane oxygen atom breaks, when a hydrogen atom adsorbs on the out-of-plane oxygen atom on the (101), but there is no bond breaking in the case of (110).

The findings from the DFT calculations suggest that the differences between the experimental results obtained for the two different materials could be determined by their different dominant surface orientation and concentration of oxygen vacancies. In the case of the IPC material the theoretical results indicate a very "clean" (101) surface. On the opposite, for the SA material one expects to have a defective (110) surface. It is very difficult to have an experimental proof for the presence of a dominant surface so, at this stage, we do not know that for sure. For the case of oxygen vacancies, we have some indications from the results of a Raman investigation performed on sensors at RT and presented in Figure 8.

The broad band of SA, which is observed between 370 and 820 cm^{-1}, consists of a series of bands, namely, the A_{1g} (633 cm^{-1}), B_{2g} (774 cm^{-1}) and E_g (475 cm^{-1}) lattice modes and three additional bands. Diéguez et al. assigned these additional bands to the IR-active A_{2u} mode (~692 cm^{-1}) and two broad bands S1 and S2, which they found between 568 and 576 and 493−542 cm^{-1}, respectively, and that were assigned to the defective surface region of the SnO_2 grains.[47] Lui et al. found that the S1 (~573 cm^{-1}) band arises from oxygen vacancies.[48] For IPC a similar broad band is not observed. Consequently, we do believe that the presence of the S1 and S2 bands indicates the presence of oxygen vacancies for the SA material.

CO Reactions with SnO_2 Surfaces. As already mentioned, CO reacts with O_O atom of the surface resulting in the formation of a CO_2 molecule and a vacancy on the SnO_2 surfaces, as described in ref 26 on the SnO_2 (101) surfaces. We found the same surface reduction also for the (110) surface with a calculated energy gain of −1.55 eV.

Adsorption/Reaction of CO in the Presence of Humidity. We studied the effect of humidity on CO detection by considering preadsorbed OH groups on the SnO_2 surfaces. The goal, here, is to find out if the reduction of the surface still occurs even in the presence of hydroxyl groups. We performed the calculations on the most stable dissociated states of water molecules on both clean and defective (110) and (101) surfaces (Figure 7c,d). We observed that the spontaneous adsorption of CO found on clean (101) and (110) surfaces are inhibited on the hydroxylated areas of (101) and (110) surfaces: a CO molecule initially placed 2 Å away from the surface as conducted conventionally in an adsorption process using the DFT study is repelled upward further from the surface to an equilibrium adsorption state where no more interaction exists. In conclusion, adsorbed hydroxyls act as a repelling species for CO adsorption on SnO_2 surfaces.

To force the reaction of CO with the hydroxylated surface, a drag method was applied on the $-z$ direction perpendicular to the surface and above the OH groups. We mainly observe the deviation of the CO molecule from hydroxylated parts of the surface to reach clean parts allowing the adsorption of CO on clean parts as described above. From a spontaneous adsorption on both tin oxide surfaces, the adsorption of CO becomes thermally activated in the presence of humidity. Activation barriers have been estimated larger than 0.40 eV.

Combining the insights obtained from the theoretical calculations for both humidity and CO one can state the following:

- In the case of the IPC material, considered to be described by the clean (101) surface, we are expecting an additive electrical effect of (1) the incipient formation of oxygen vacancies, because of the reaction with water vapor, and (2) the formation of oxygen vacancies, which takes place on the nonhydroxylated parts, because of the reduction with CO. In the presence of humidity the effect of CO will be reduced, because there will be less "free" SnO_2 surface to react with. The resistance will be lower in the presence of both gases, because both of them are having an electrical effect.
- In the case of the SA material, considered to be described by the defective (110) surface, we are expecting that the reaction with water vapor will not change the electrical resistance because the process of healing oxygen vacancies by filling them with hydroxyl groups is neutral. The exposure to CO will result in a lower concentration of oxygen vacancies in the presence of humidity due to the decrease of the reactive "free" SnO_2 surface. The total resistance in the presence of both humidity and CO will be lower that in the presence of CO only.

■ CONCLUSIONS

On the basis of this work the current understanding of water vapor impact on SnO_2, namely, humidity behaving as a reducing gas, has to be changed to include the concept that humidity can also act in an electroneutral way. A new reaction mechanism was proposed, in which two rooted hydroxyl groups can be formed by H_2O adsorption on lattice oxygen and an oxygen vacancy. Furthermore, it was shown that the way in which humidity reacts on the respective SnO_2 surface also affects its cross-interference in the detection of CO. Because the two materials chosen for our study are describing the extreme cases we found in our extended SnO_2 investigation, it is to be expected that in most cases both types of surface reactions coexist and it is even possible that depending on the ambient conditions their respective weights change. Our findings are also opening up intriguing synthesis strategy avenues, because the theoretical studies suggest that this different type of electrical effects is associated with different surfaces and different concentrations of oxygen vacancies. Furthermore, our results suggest that the role of oxygen vacancies is not fully understood and complete the recent findings for WO_3, where we even observed the oxidation of the material in the presence of humidity.[49]

Our attempt to explain the effect of humidity is also an example of what can be achieved by combining advanced operando investigations with theoretical calculations. The experiments are pointing the theory to the relevant tasks, in our case the finding of surface reactions that are electroneutral, whereas the theory highlights the weaknesses of the modeling only on the basis of experiments, in our case the absence of a full reduction of the surface by water vapor even in the case of the (101) clean surface. However, even if the combination of

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those both approaches is a very powerful investigation tool it does not provide a definitive proof for the correctness of the proposed models. For that to happen there is still need for additional experimental work, which could be provided by studies on single crystalline or epitaxial layers. These types of model systems will need to be thoroughly studied, in operando conditions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.7b06253.

Additional DRIFT and XRD spectroscopic data, SEM pictures, and electrical resistance measurements are provided (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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